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Software Components of a DBMS

Technical Presentation

CSA0506 :Database Management Systems

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Introduction

- A **Database Management System (DBMS)** is software used to create, store, organize, and manage data efficiently.
- It acts as an interface between users, applications, and the database.
- DBMS allows users to **insert, update, delete, and retrieve** data easily.
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Storage Manager

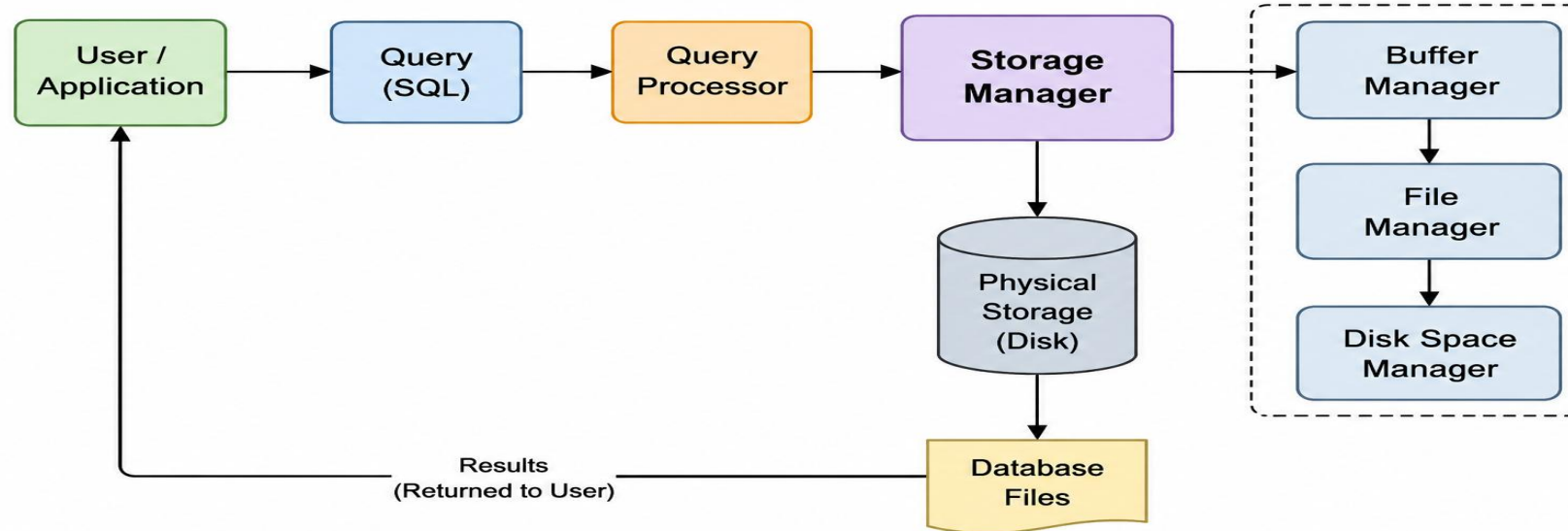
- The **Storage Manager** is responsible for storing, retrieving, and updating data in the database.
- It acts as a bridge between the database and the physical storage devices (hard disk or SSD).
- It organizes data into files and pages for efficient access.

Example:

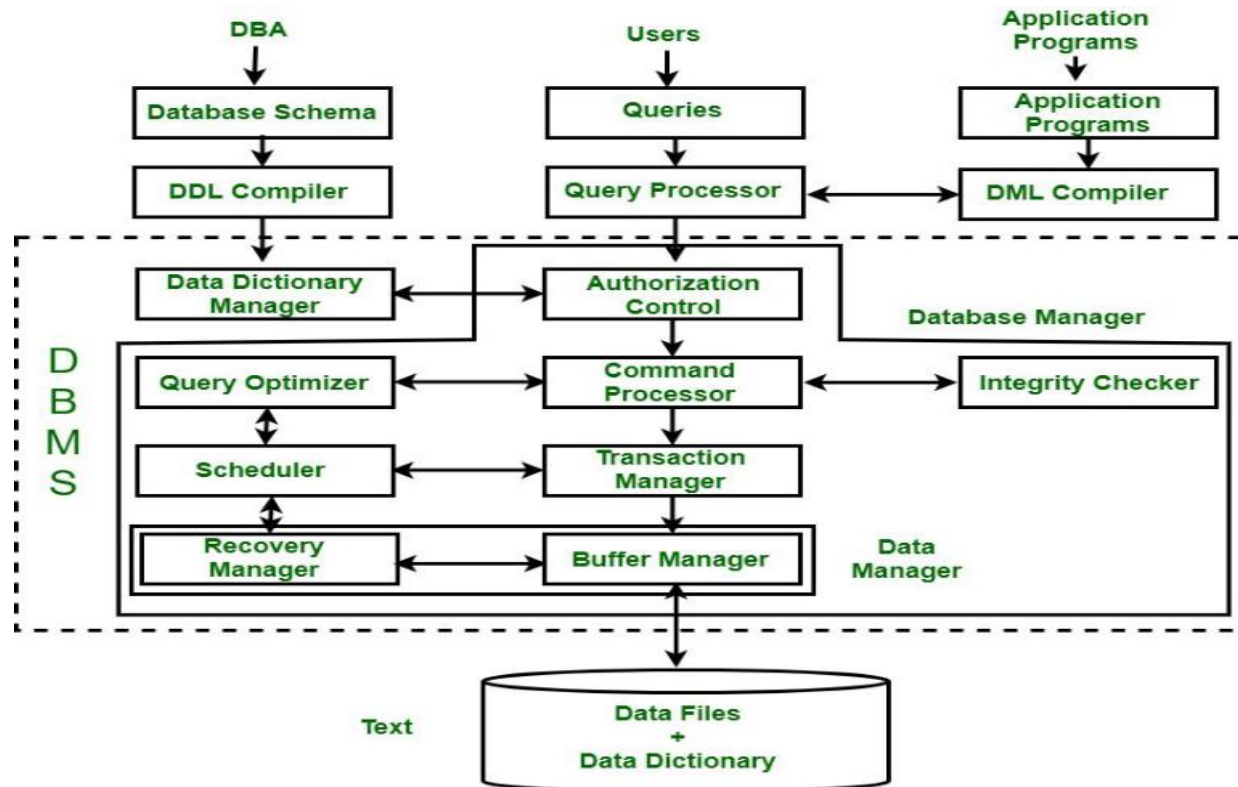
When a student record is saved in a college database, the Storage Manager decides where and how the record is stored.

Storage Flow Chat

Storage Manager in DBMS



Architecture daigram





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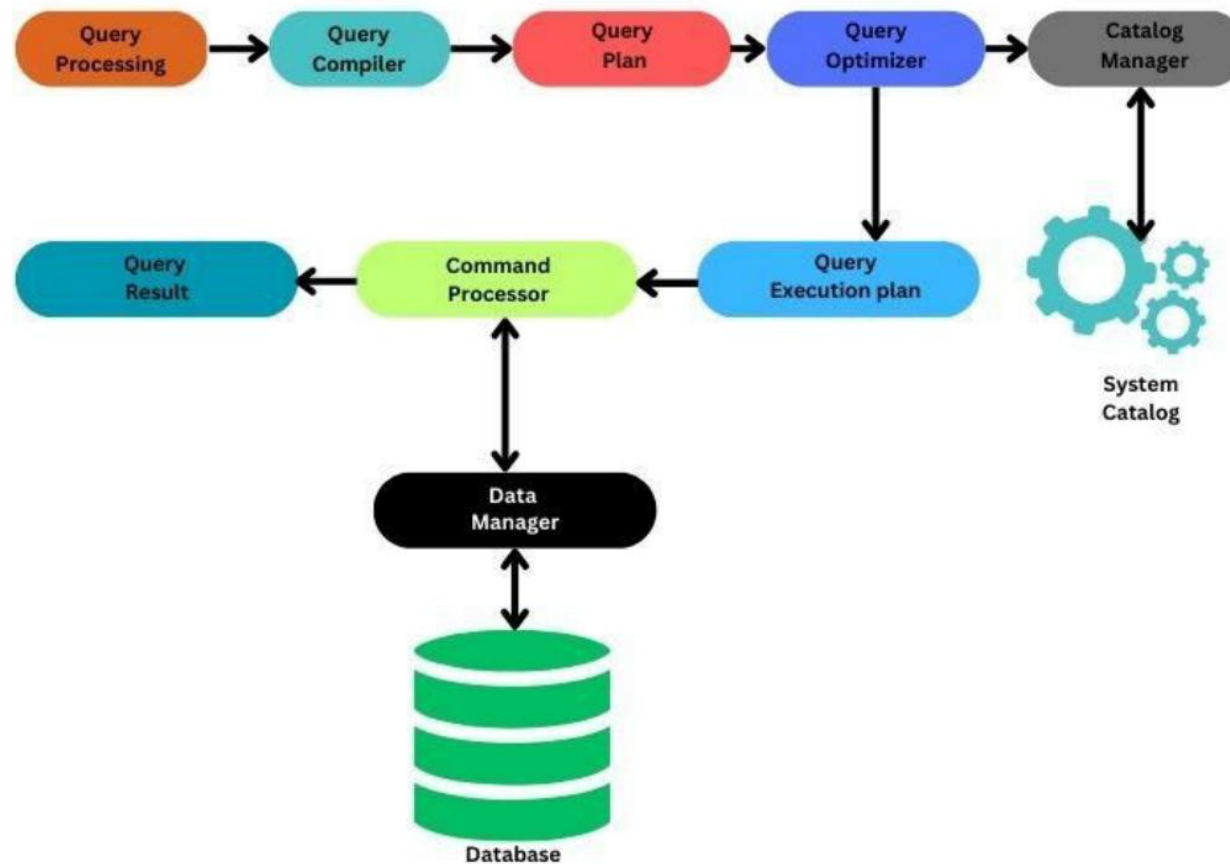
Query Processor

- The **Query Processor** interprets and executes SQL commands submitted by users.
- It checks the query for syntax and logical errors.
- It creates an optimized execution plan to retrieve data quickly.

Example:

When you run `SELECT * FROM Students;`, the Query Processor processes the command and returns the student records.

Query Processor Flow





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Future scope

Cloud Database Management

- More organizations are moving databases to cloud platforms.
- Cloud DBMS provides scalability, flexibility, and lower maintenance costs.

Artificial Intelligence (AI) Integration

- AI helps automate database management tasks.
- It improves query optimization, performance tuning, and error detection



Thank You!!!